

## Tier Architecture with Load Balancer

### Step 1: Create the VPC

1. Sign in to the **AWS Management Console** and set the Region in the top-right (choose where you want the VPC).
2. From **Services** → open **VPC**. In the VPC dashboard left menu choose **Your VPCs**. Click **Create VPC** (button).
3. Choose **Create VPC** (simple form):
  - **Name tag:** ThreeTierVPC
  - **IPv4 CIDR block:** 10.0.0.0/16
  - **IPv6 CIDR block:** No (leave disabled)
  - **Tenancy:** Default
4. Click **Create VPC**.

The screenshot shows the AWS Management Console 'Create VPC' page. The 'VPC settings' section is expanded, showing 'Resources to create' with 'VPC only' selected. The 'Name tag - optional' field is filled with 'ThreeTierVPC'. Under 'IPv4 CIDR block', 'IPv4 CIDR manual input' is selected, and the 'IPv4 CIDR' field contains '10.0.0.0/16'. Under 'IPv6 CIDR block', 'No IPv6 CIDR block' is selected. The 'Tenancy' dropdown is set to 'Default'. The 'Tags' section shows a key-value pair 'ThreeTierVPC'. At the bottom right, there are buttons for 'Cancel', 'Preview code', and 'Create VPC'.

5. After creation, click **Close**. In **Your VPCs** you should now see the ThreeTierVPC entry with a VPC ID (vpc-0ac41ed0700b54e18).

## Step 2: Create Subnets

1. Open **Services** → **VPC** → in the left menu click **Subnets**.
2. Click **Create subnet**.
3. Fill the form for the **Public Subnet (Web)**:
  - **Name tag**: Public-Subnet
  - **VPC**: select ThreeTierVPC
  - **Availability Zone**: pick an AZ (e.g., us-east-1a)
  - **IPv4 CIDR block**: 10.0.1.0/24
  - (Leave IPv6 blank)
  - Click **Create subnet**.

The screenshot shows the AWS 'Create subnet' interface. The 'VPC' section shows 'vpc-dac41ed700d54e18 (ThreeTierVPC)' selected. Under 'Subnet settings', the 'Subnet name' is 'Public-Subnet', the 'Availability Zone' is 'Asia Pacific (Mumbai) / ap-south-1a', the 'IPv4 VPC CIDR block' is '10.0.0.0/16', and the 'IPv4 subnet CIDR block' is '10.0.1.0/24'. A tag with key 'Name' and value 'Public-Subnet' is added. The 'Create subnet' button is at the bottom right.

4. Repeat **Create subnet** for **Private Subnet A (App)**:
  - Name: App-Subnet
  - VPC: ThreeTierVPC
  - AZ: choose a different AZ if you want AZ separation (e.g., us-east-1b)
  - IPv4 CIDR: 10.0.2.0/24
  - Create.

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**Create subnet**

**VPC**  
VPC ID: vpc-dac41ed700b54e18 (ThreeTierVPC)  
Associated VPC CIDRs: 10.0.0/16

**Subnet settings**  
Specify the CIDR blocks and Availability Zone for the subnet.

**Subnet 1 of 1**  
Subnet name: App-Subnet  
Availability Zone: Asia Pacific (Mumbai) / ap-south-1c  
IPv4 VPC CIDR block: 10.0.0/16  
IPv4 subnet CIDR block: 10.0.0/24  
Tags - optional: Key: Name, Value: App-Subnet

### 5. Repeat **Create subnet** for **Private Subnet B (DB)**:

- Name: DB-Subnet
- VPC: ThreeTierVPC
- AZ: choose us-east-1c
- IPv4 CIDR: 10.0.3.0/24
- Create.

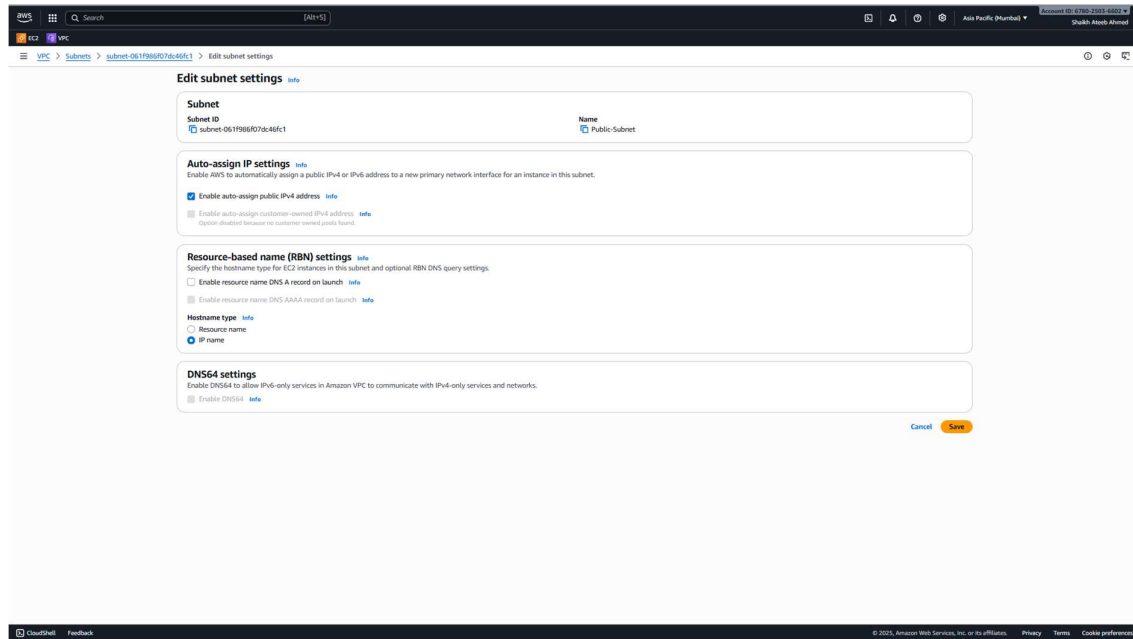
**Create subnet**

**VPC**  
VPC ID: vpc-dac41ed700b54e18 (ThreeTierVPC)  
Associated VPC CIDRs: 10.0.0/16

**Subnet settings**  
Specify the CIDR blocks and Availability Zone for the subnet.

**Subnet 1 of 1**  
Subnet name: DB-Subnet  
Availability Zone: Asia Pacific (Mumbai) / ap-south-1c  
IPv4 VPC CIDR block: 10.0.0/16  
IPv4 subnet CIDR block: 10.0.3.0/24  
Tags - optional: Key: Name, Value: DB-Subnet

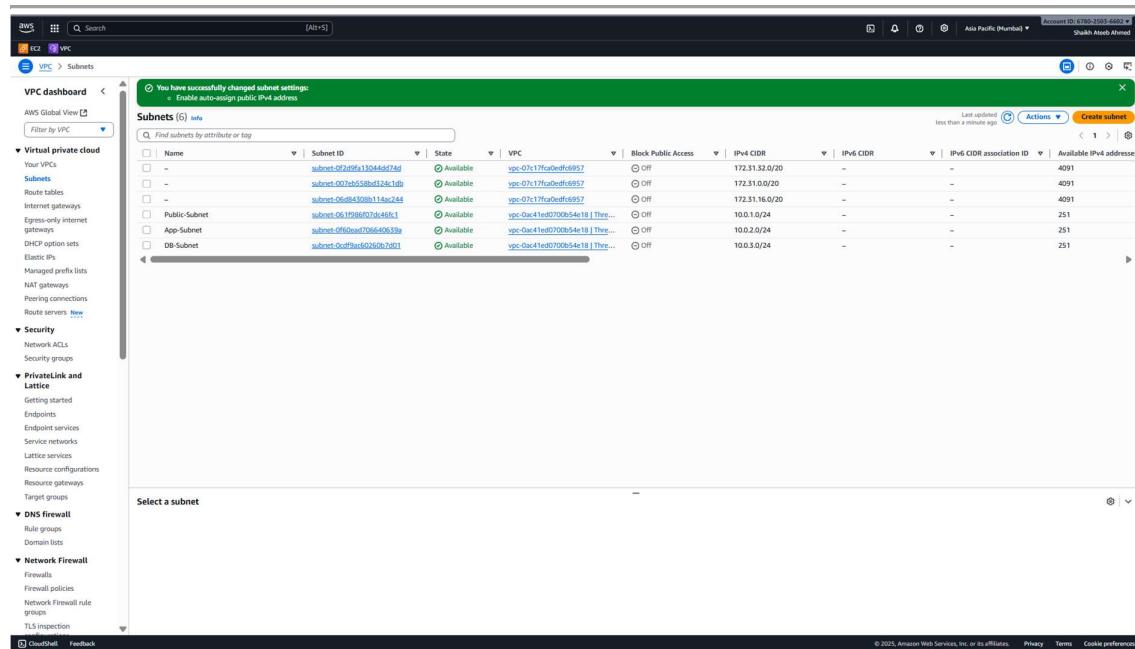
6. Configure auto-assign public IP for the Public Subnet so instances launched there can receive a public IP:
  - In **Subnets** list select Public-Subnet → **Actions** → **Edit subnet settings** → check **Enable auto-assign public IPv4 address** → **Save changes**.



7. Verify:
  - In **Subnets** view you should see three entries with CIDRs 10.0.1.0/24, 10.0.2.0/24, 10.0.3.0/24 and the Name tags you set.
  - Click each subnet to view the **Subnet details** and confirm it's associated with ThreeTierVPC.

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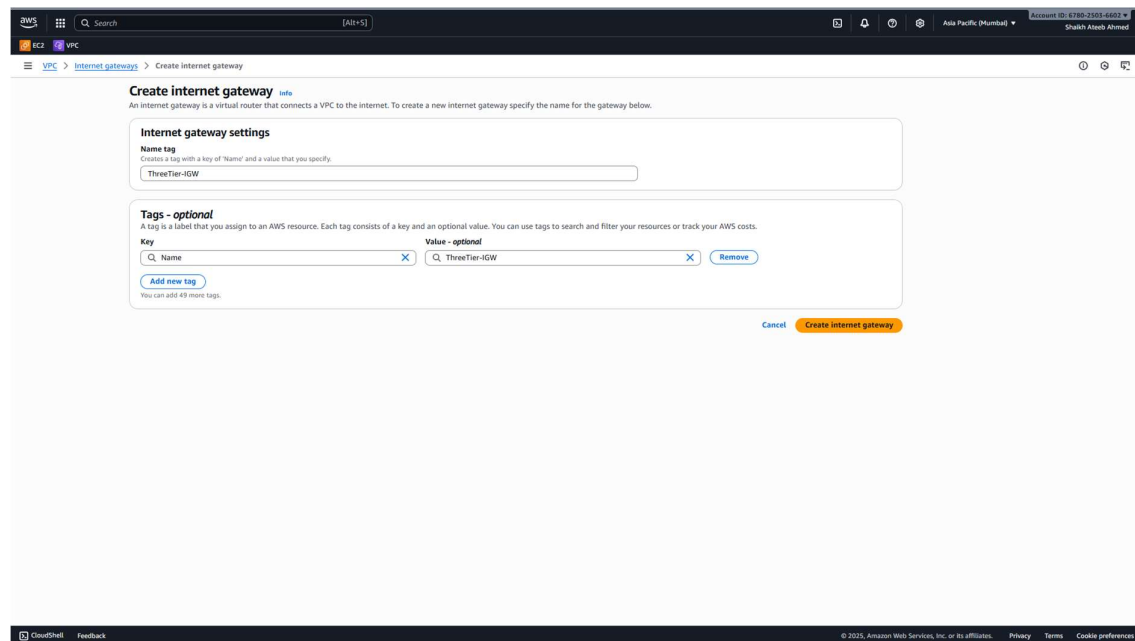
## 25-08-2025



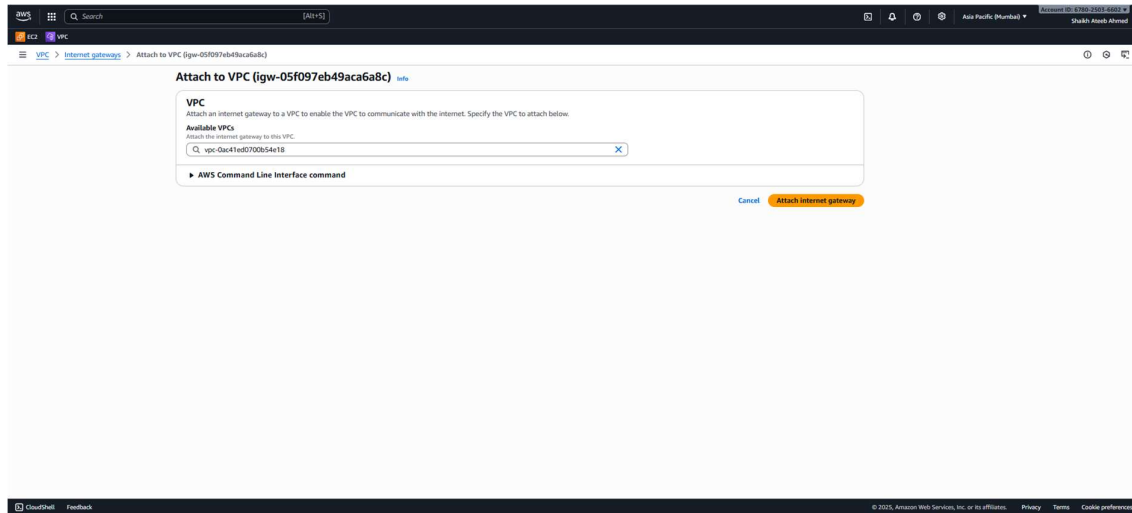
### Step 3: Internet Gateway & NAT Gateway

#### 1) Create and attach an Internet Gateway (IGW)

1. Sign in to the AWS Console → **Services** → **VPC**.
2. On the left menu click **Internet Gateways** → **Create internet gateway**.
3. Give it a **Name tag**: ThreeTier-IGW → **Create internet gateway**.



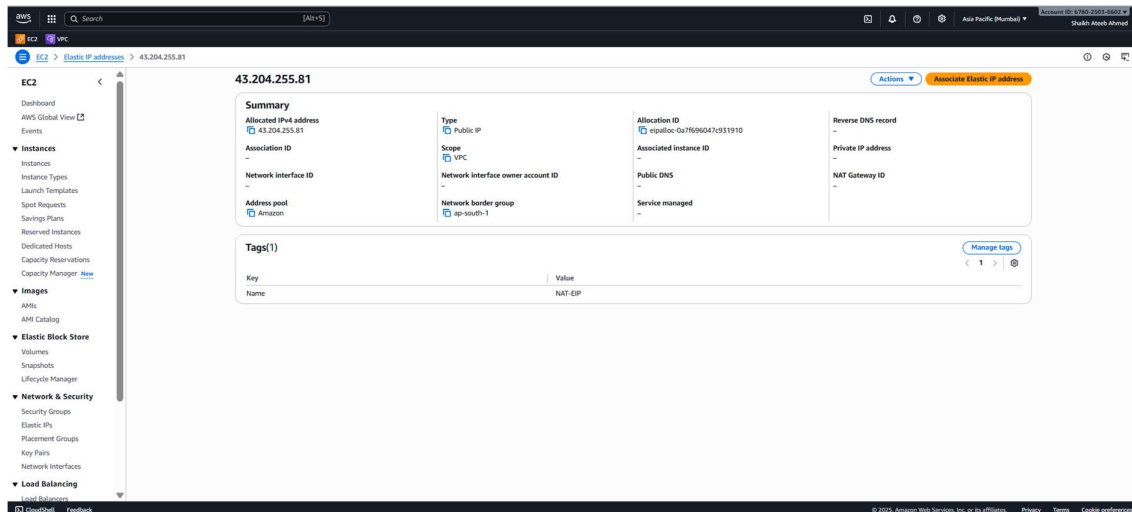
4. Select the created IGW (ThreeTier-IGW) → **Actions** → **Attach to VPC**.
5. Choose **ThreeTierVPC** → **Attach internet gateway**.



6. Verify: select the IGW and check the **Attachment** shows ThreeTierVPC (state: attached).

## 2) Allocate an Elastic IP (EIP) for the NAT Gateway

1. Services → **EC2** → left menu → **Elastic IPs** → **Allocate Elastic IP address**.
2. Choose **Amazon's pool of IPv4 addresses** → **Allocate**.
3. Tag it (optional) with Name: NAT-EIP.



4. Note the **Allocation ID** (eipalloc-0a7f696047c931910) or the Elastic IP itself.

### 3) Create a NAT Gateway in the Public Subnet

1. Services → **VPC** → left menu → **NAT Gateways** → **Create NAT gateway**.
2. Settings:
  - **Name tag:** ThreeTier-NAT
  - **Subnet:** choose Public-Subnet (must be a public subnet with a route to the IGW)
  - **Elastic IP allocation ID:** choose the EIP you allocated

The screenshot shows the AWS Management Console interface for creating a NAT gateway. The 'NAT gateway settings' section includes a 'Name' field with the value 'ThreeTier-NAT', a 'Subnet' dropdown menu showing 'subnet-061f989f78d48c1 Public-Subnet', and a 'Connectivity type' section with 'Public' selected. Below this is the 'Elastic IP allocation ID' dropdown menu and an 'Allocate Elastic IP' button. The 'Additional settings' section includes a 'Tags' section where a tag with key 'Name' and value 'ThreeTier-NAT' is added. The bottom right of the form has 'Cancel' and 'Create NAT gateway' buttons.

3. Click **Create NAT gateway**.
4. NAT status will appear in the console (pending → available). Once **Available**, you can use it in route tables.

**Verify:** Select the NAT Gateway and confirm:

- Subnet = Public-Subnet
- Elastic IP listed
- State = Available

### Step 4: Route Tables

#### 1) Create Public Route Table and attach IGW

1. **AWS Console** → **Services** → **VPC** → left menu → **Route Tables**.

2. Click Create route table.

3. Set:

- Name tag: Public-RT
- VPC: select ThreeTierVPC

The screenshot shows the AWS Management Console interface for creating a new route table. The page title is 'Create route table'. Below the title, there is a brief description: 'A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.' The form is divided into two main sections: 'Route table settings' and 'Tags'. In the 'Route table settings' section, the 'Name' field is set to 'Public-RT' and the 'VPC' dropdown is set to 'vpc-Quc41ed0700b64e18 (ThreeTierVPC)'. In the 'Tags' section, a tag is added with the key 'Name' and the value 'Public-RT'. At the bottom right of the form, there are two buttons: 'Cancel' and 'Create route table' (which is highlighted in orange).

4. Click Create route table.

5. Select the newly created Public-RT from the list.

6. In the Routes tab → click Edit routes → Add route:

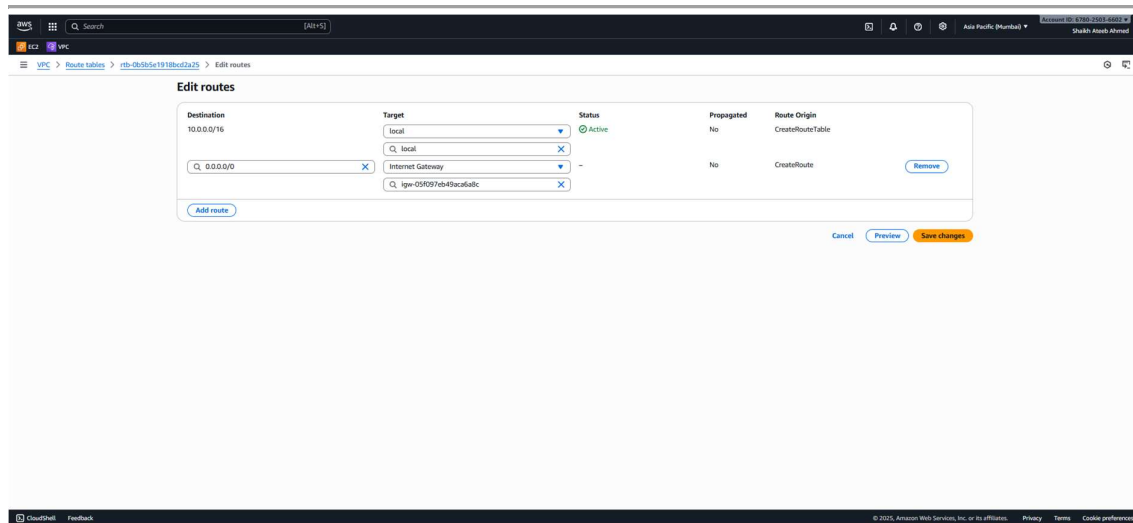
- Destination: 0.0.0.0/0
- Target: select Internet gateway and choose ThreeTier-IGW

7. Save routes.

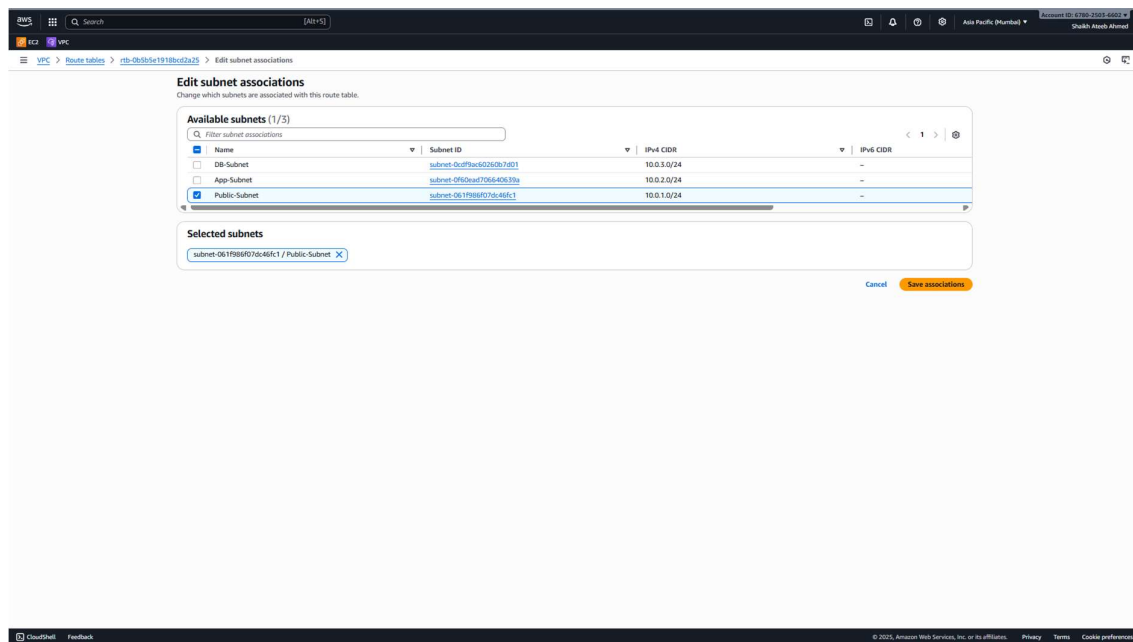


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8. Switch to Subnet associations tab → Edit subnet associations → check Public-Subnet → Save.



9. Verify the Public-RT now shows the route 0.0.0.0/0 -> igw-xxxxx and is associated with Public-Subnet.

## 2) Create Private Route Table for App Subnet (uses NAT)

1. In Route Tables → Create route table.

2. Set:

- Name tag: App-Private-RT

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### ○ VPC: ThreeTierVPC

**Create route table**

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

**Route table settings**

**Name - optional**  
Create a tag with a key of 'Name' and a value that you specify.  
App-Private-RT

**VPC**  
The VPC to use for this route table.  
vpc-0a41ed0700b64e18 (ThreeTierVPC)

**Tags**  
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

**Key**  
Name

**Value - optional**  
App-Private-RT

[Add new tag](#)  
You can add 49 more tags.

[Cancel](#) [Create route table](#)

3. Click Create route table.

4. Select App-Private-RT → Routes → Edit routes → Add route:

○ Destination: 0.0.0.0/0

○ Target: select NAT gateway → choose ThreeTier-NAT

**Edit routes**

| Destination | Target      | Status | Propagated | Route Origin     |
|-------------|-------------|--------|------------|------------------|
| 10.0.0.0/16 | local       | Active | No         | CreateRouteTable |
| 0.0.0.0/0   | NAT Gateway | -      | No         | CreateRoute      |

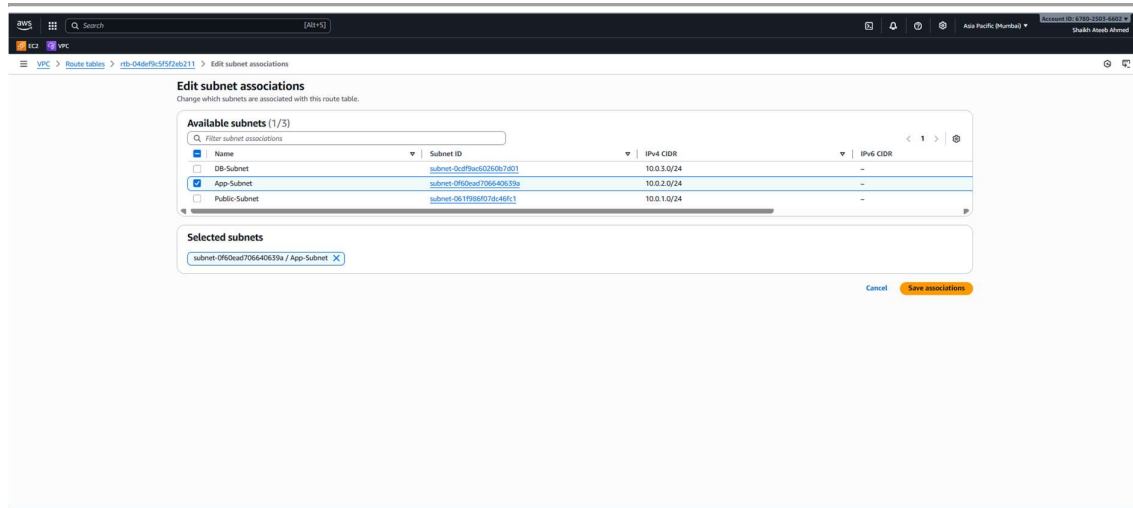
[Add route](#) [Cancel](#) [Preview](#) [Save changes](#)

5. Save routes.

6. Go to Subnet associations → Edit subnet associations → check App-Subnet → Save.

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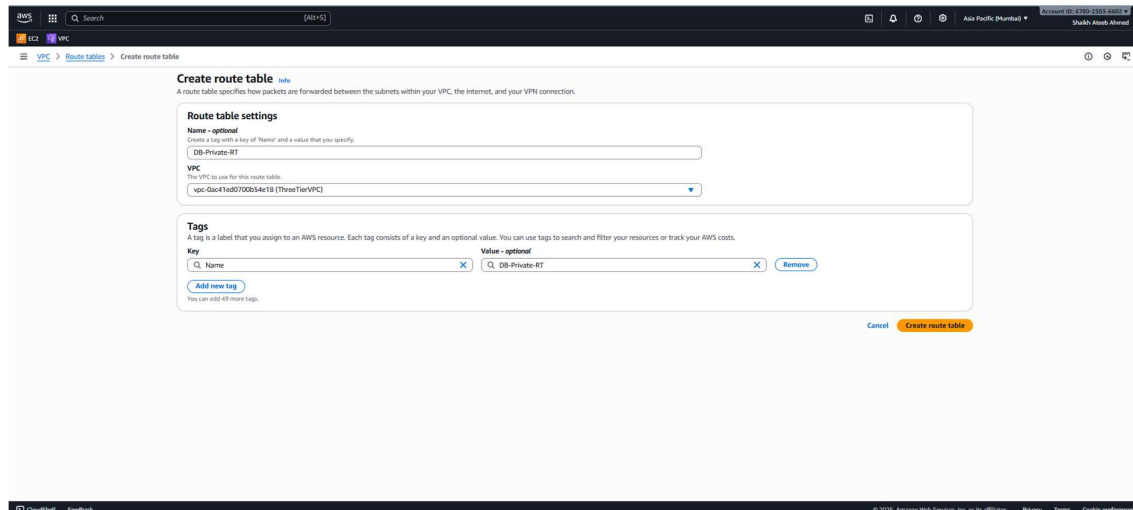
7. Verify App-Private-RT shows 0.0.0.0/0 -> nat-xxxxx and associated with App-Subnet.

### 3) Create Private Route Table for DB Subnet (no Internet)

1. In Route Tables → Create route table.

2. Set:

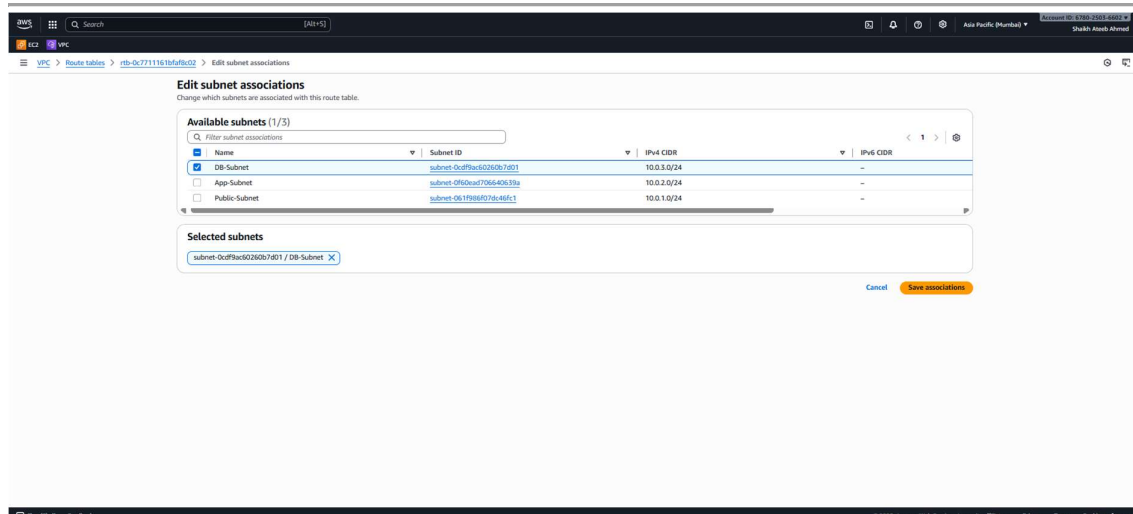
- Name tag: DB-Private-RT
- VPC: ThreeTierVPC



3. Click Create route table.

4. Select DB-Private-RT → Routes tab: you should see only the local route (e.g., 10.0.0.0/16 -> local). Do not add a 0.0.0.0/0 route.

5. Subnet associations → Edit subnet associations → check DB-Subnet → Save.



6. Verify: DB-Private-RT shows only the local route and is associated with DB-Subnet.

### Step 5: Security Groups (SGs)

1. Sign in → Services → EC2 → left menu → Security Groups → Create security group.

#### Create Web-SG

- **Name:** Web-SG
- **VPC:** ThreeTierVPC
- **Description:** Web servers (HTTP & SSH)
- **Inbound rules:**
  - Type: **HTTP**, Port: **80**, Source: **Anywhere 0.0.0.0/0**
  - Type: **SSH**, Port: **22**, Source: **My IP** (click the **My IP** option in the console)
- **Outbound:** default **Allow all** (leave as-is)

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**Create security group**

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

**Basic details**

Security group name:

Description:

VPC:

**Inbound rules**

| Type  | Protocol | Port range | Source   | Description - optional |
|-------|----------|------------|----------|------------------------|
| http  | TCP      | 80         | Anywhere | 0.0.0.0/0              |
| https | TCP      | 443        | My IP    | 192.168.1.1/16         |

**Outbound rules**

| Type        | Protocol | Port range | Destination | Description - optional |
|-------------|----------|------------|-------------|------------------------|
| All traffic | All      | All        | Custom      | 0.0.0.0/0              |

**Tags - optional**

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. We tag resources with the resource.

[Add new tag](#)

[Cancel](#) [Create security group](#)

- Create.

Tip: For initial testing you may set HTTP = 0.0.0.0/0, then later update to allow only ALB-SG as source.

## Create App-SG

- **Name:** App-SG
- **VPC:** ThreeTierVPC
- **Description:** App server (accepts traffic from Web-SG)
- **Inbound rules:**
  - Type: **Custom TCP**, Port: **8080**, Source: **Security group** → choose Web-SG (select by name/id)
- **Outbound:** default **Allow all**

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**Create security group**

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

**Basic details**

**Security group name** info  
App-SG  
Name cannot be edited after creation.

**Description** info  
App server (accepts traffic from Web-SG)

**VPC** info  
vpc-0a7f4d705554b18 (ThreeTierVPC)

**Inbound rules** info

| Type       | Protocol | Port range | Source              | Description - optional |
|------------|----------|------------|---------------------|------------------------|
| Custom TCP | TCP      | 8080       | sg-02b4d0f02388b0a8 | sg-02b4d0f02388b0a8    |

**Outbound rules** info

| Type        | Protocol | Port range | Destination | Description - optional |
|-------------|----------|------------|-------------|------------------------|
| All traffic | All      | All        | 0.0.0.0/0   |                        |

Rules with destination of 0.0.0.0/0 or ::0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific, known IP addresses.

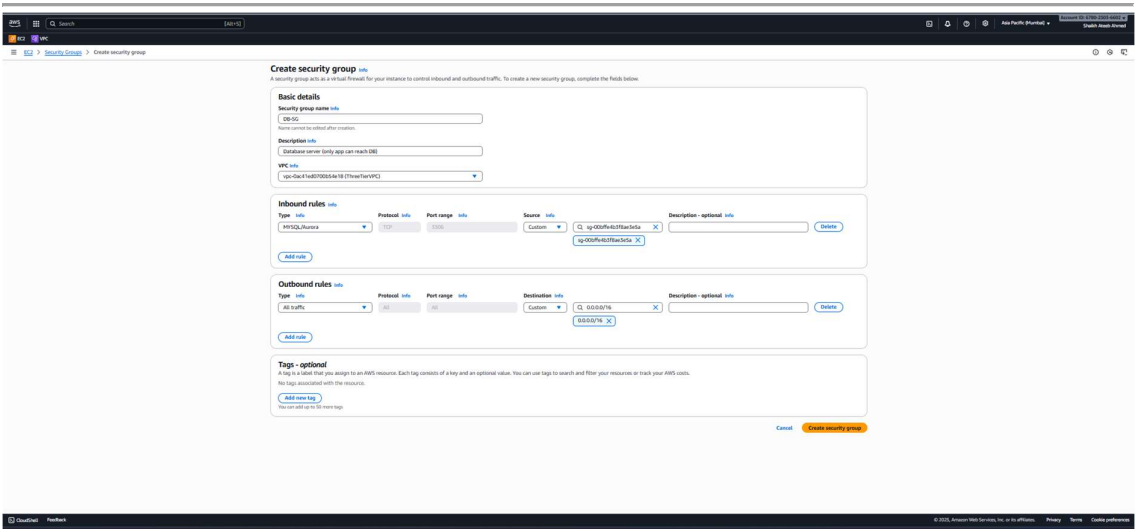
**Tags - optional**  
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. No tags associated with the resource.

**Create security group**

- Create.

## Create DB-SG

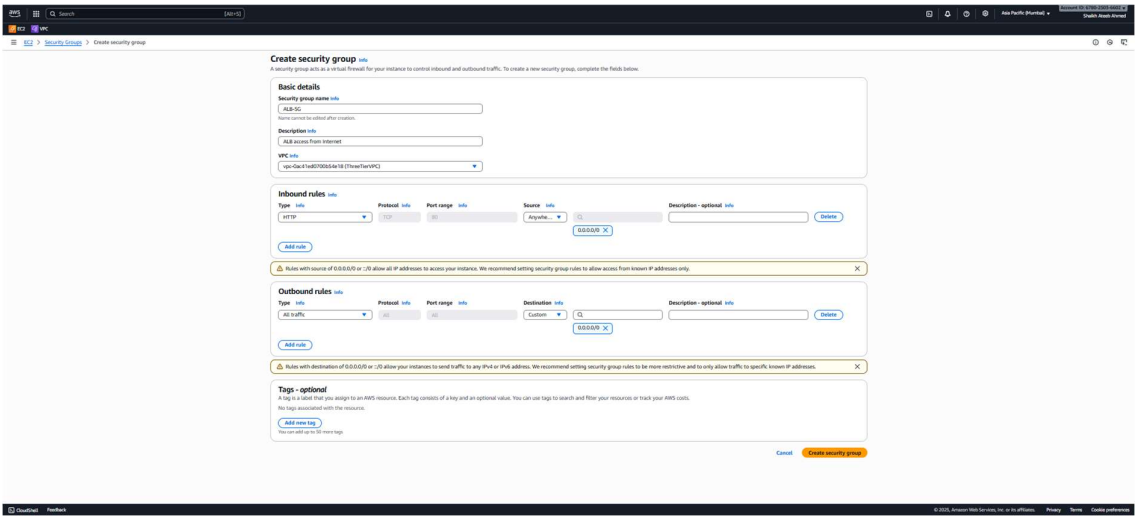
- **Name:** DB-SG
- **VPC:** ThreeTierVPC
- **Description:** Database server (only app can reach DB)
- **Inbound rules:**
  - Type: choose **MySQL/Aurora** port **3306**, Source: **Security group** → choose App-SG
- **Outbound:** change from default **Allow all** to **Restrict to local VPC:**
  - In the Security Group → **Outbound rules** → **Edit outbound rules** → Remove the default rule (0.0.0.0/0).
  - Add new Outbound rule: **All traffic** (or required ports) → Destination: **Custom** → enter your VPC CIDR 10.0.0.0/16.



- Create.

Create ALB-SG

- **Name:** ALB-SG
- **VPC:** ThreeTierVPC
- **Description:** ALB access from Internet
- **Inbound rules:**
  - Type: **HTTP**, Port: **80**, Source: **Anywhere 0.0.0.0/0** (and ::/0 for IPv6 if used)
- **Outbound:** default **Allow all** (so ALB can reach targets)



- Create.

## **Step 6: Launch EC2 Instances**

### **1) Launch Web Servers (2 instances in Public-Subnet)**

1. EC2 → **Instances** → **Launch instances**.
2. Choose an AMI: **Amazon Linux 2 (x86\_64)** or Ubuntu.
3. Instance type: t3.micro (or t2.micro).
4. Configure Instance:
  - Network: ThreeTierVPC
  - Subnet: Public-Subnet
  - Auto-assign Public IP: **Enable**
  - IAM role: optional (for SSM use)
5. Add storage: default ok.
6. Add tags: Name=Web-1 (you'll create second after or launch two with different tags).
7. Configure Security Group: choose Web-SG.
8. Key pair: select the key pair that u have created.
9. Advanced → **User data**: paste this (Amazon Linux 2 example for Web-1):

```
#!/bin/bash
```

```
yum update -y
```

```
yum install -y httpd
```

```
systemctl enable httpd
```

```
systemctl start httpd
```

```
echo "Hello from Web Server 1 (INSTANCE_ID: $(curl -s http://169.254.169.254/latest/meta-data/instance-id))" > /var/www/html/index.html
```



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[illegible]

SMB-CG

[Launch an instance](#)

### ▼ Network settings [info](#)

**VPC:** [required](#) [info](#)

sg-0d4f7ad0700304a18 (ElasticNet VPC)  
Subnet ID

**Address:** [info](#)

|                                                                                                                            |               |
|----------------------------------------------------------------------------------------------------------------------------|---------------|
| subnet-081996070a46451<br>Amazon Linux 2023 AMI (Amazon Linux 2)                                                           | Public Subnet |
| Subnet ID: subnet-081996070a46451 Availability Zone: us-east-1c (us-east-1) Zone type: Availability Zone CIDR: 10.0.0.0/28 |               |

[Create new subnet](#)

**Auto assign public IP:** [info](#)

Select

**ElasticNet (security group):** [info](#)

Security groups that control the traffic to your instance. Select one or more security groups to apply specific traffic to each your instance.

[Create security group](#) [Select existing security group](#)

**Compare security groups:** [info](#)

Select security groups

W00-SG sg-02da9e97923880a05 X  
No IAM role associated here

[Compare security group rules](#)

**Advanced network configuration**

### ▼ Configure storage [info](#)

To x GB gp3 Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

[Click refresh to view backup information](#)

The steps that you specify determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 File systems **EB**

### ▼ Advanced details [info](#)

**Default java directory:** [info](#)

Select [Create new directory](#)

**JMX Instance profile:** [info](#)

Select [Create new JMX profile](#)

**Instance type:** [info](#)

m5.xlarge

**DNS Hostname:** [info](#)

- ☒ Enable IPv6 (to request DNS requests)
- ☐ Enable resource-based IPv6 (to request DNS requests)
- ☐ Enable resource-based IPv6 (AAA record) (DNS requests)

**Instance auto-recovery:** [info](#)

Select

**Shutdown behavior:** [info](#)

Select

**Stop-AfterMaintenance behavior:** [info](#)

Select

### ▼ Summary

Number of instances: [info](#)

1

**Software Usage (AMI)**

Amazon Linux 2023 AMI 2023.0.2..x86\_64  
ami-0a0070174eac00000

**Virtual server type (instance type)**

t3.xlarge

**ElasticNet (security group)**

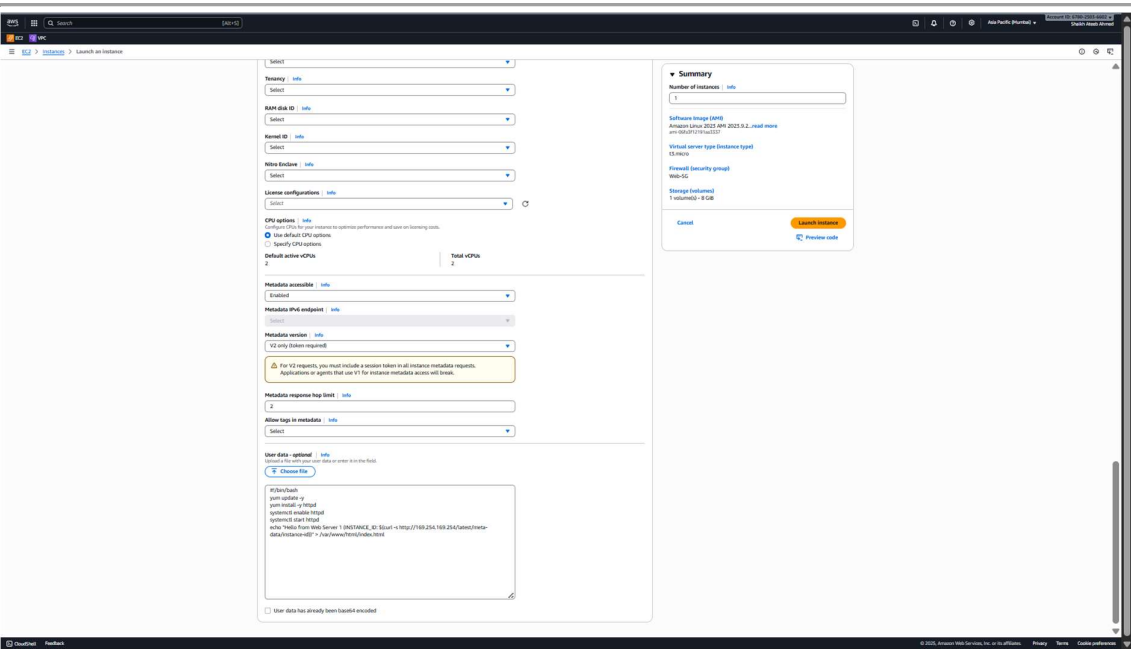
W00-SG

**Storage (instances)**

1 volumes: 0 GB

[Cancel](#) [Launch instance](#) [Previous step](#)

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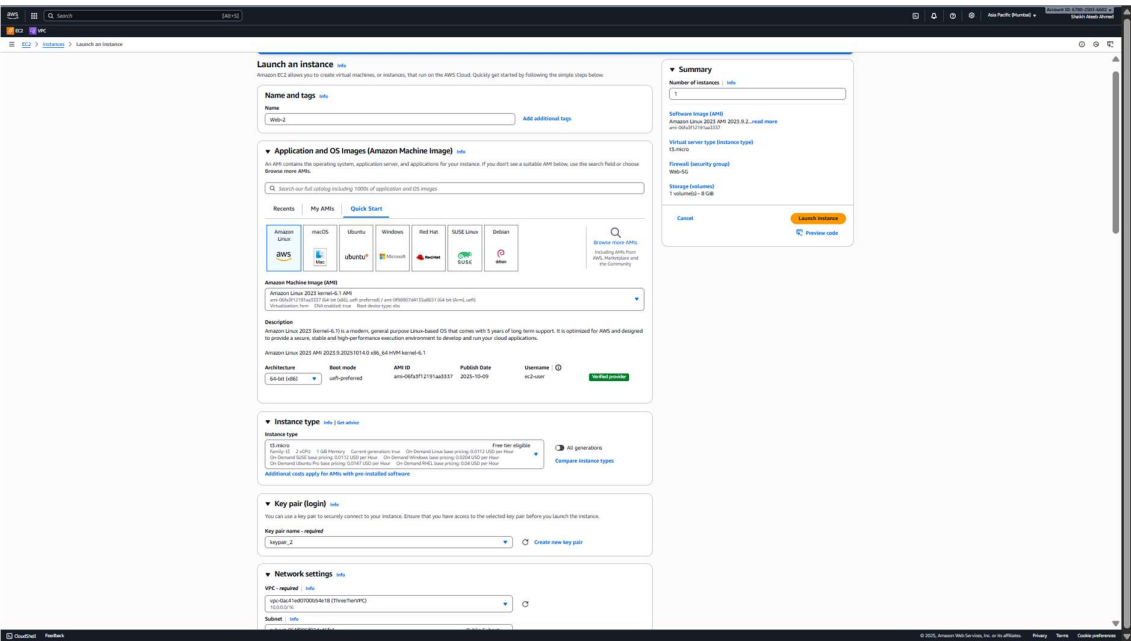


10. Launch instance → Wait until instance state = **running**. Note its Public IPv4 and Public DNS.

11. Repeat to create **Web-2** — change the user-data message to:

```
echo "Hello from Web Server 2 (HOSTNAME: $(hostname))" > /var/www/html/index.html
```

(or launch two instances in one go and modify tags after).



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The image displays two screenshots of the AWS Management Console, specifically the 'Launch instances' wizard. The first screenshot shows the 'Key pair (login)', 'Network settings', 'Configure storage', and 'Advanced details' sections. The second screenshot shows the 'Summary' section and the 'Launch instances' button.

**Key pair (login)**

You can use a key pair to securely connect to your instances. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name:  [Create new key pair](#)

**Network settings**

VPC:  [Create new VPC](#)

Subnet:  [Create new subnet](#)

Auto-assign public IP:  [Info](#)

Firewall (security group):  [Select existing security group](#)

Common security groups:  [Compare security group rules](#)

**Configure storage**

Root volume:  [Add new volume](#)

**Advanced details**

Domain join directory:  [Create new directory](#)

IAM instance profile:  [Create new IAM profile](#)

Instance type:  [Info](#)

OS:  [Info](#)

OS options:  [Info](#)

OS options:  [Info](#)

Default instance profile:  [Info](#)

Instance auto-recovery:  [Info](#)

**Summary**

Number of instances:  [Info](#)

Software image (AMI):  [Info](#)

Virtual server type (instance type):  [Info](#)

Firewall (security group):  [Info](#)

Storage (root volume):  [Info](#)

[Launch instances](#) [Previous code](#)

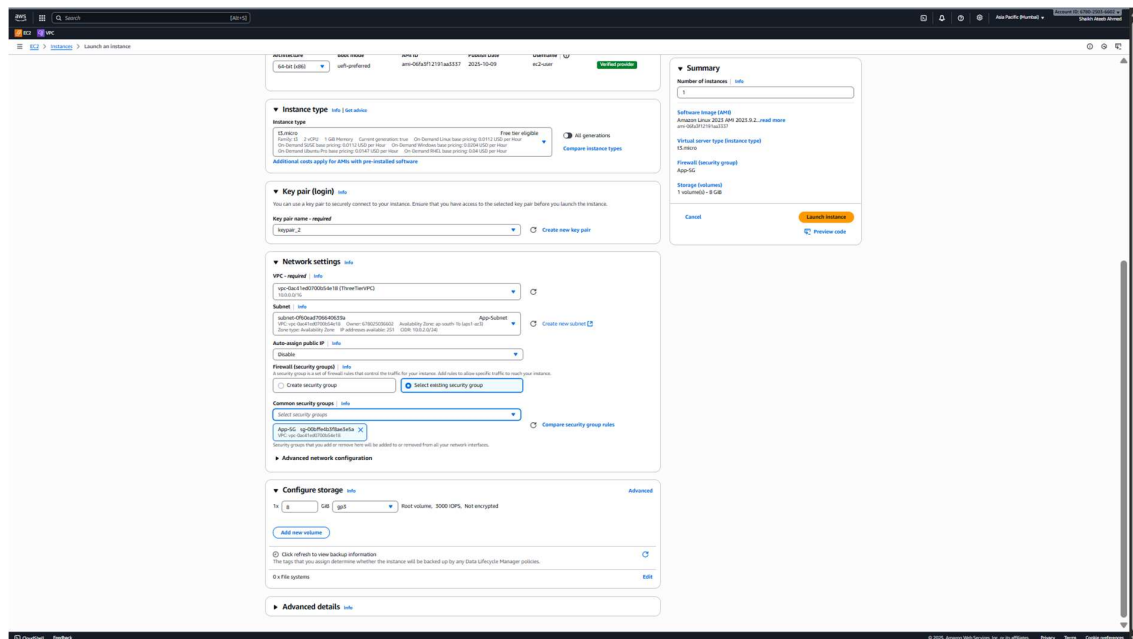
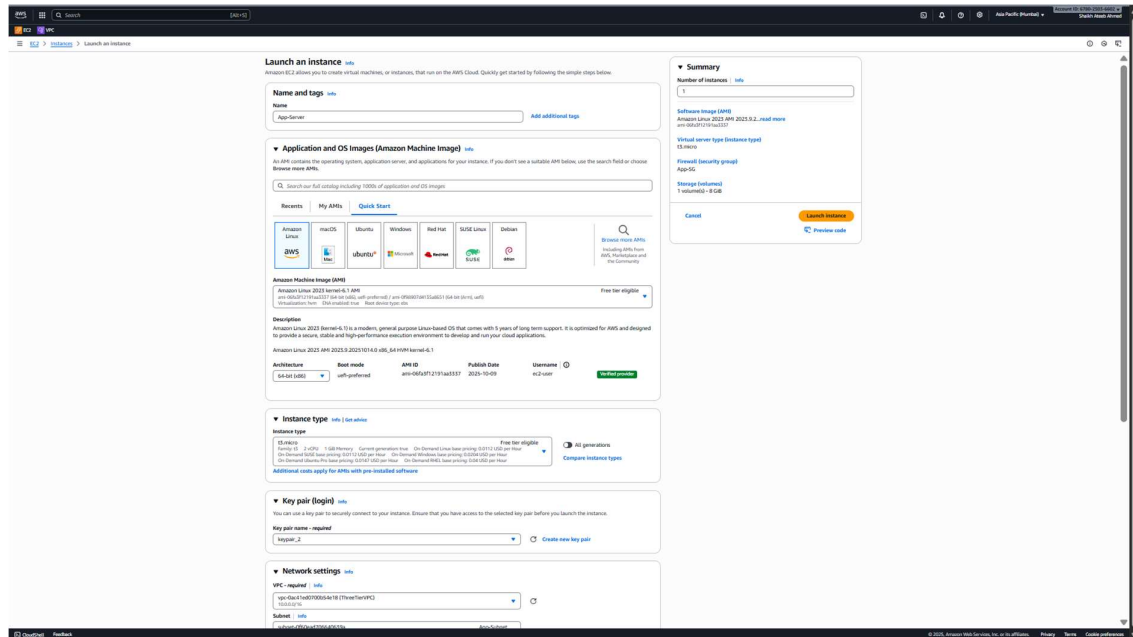
## 2) Launch App Server (Private subnet, no public IP)

1. EC2 → **Instances** → **Launch instances**.
2. Choose an AMI: **Amazon Linux 2 (x86\_64)** or Ubuntu.
3. Instance type: t3.micro (or t2.micro).
4. Configure Instance:
  - o Network: ThreeTierVPC

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- Subnet: **App-Subnet**
  - Auto-assign Public IP: **Disable**
5. Add Tags: Name=App-Server.
6. Security Group: assign App-SG.



6. Launch. Confirm it has **no public IP** in the instance description.

**3) Launch DB Server (Private, no public IP, no internet)**

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1. EC2 → **Instances** → **Launch instances**.
2. Choose an AML: **Amazon Linux 2 (x86\_64)** or Ubuntu.
3. Instance type: t3.micro (or t2.micro).
4. Configure Instance:
  - Subnet: **DB-Subnet**
  - Auto-assign Public IP: **Disable**
5. Add Tags: Name=DB-Server.
6. Security Group: assign DB-SG.

[illegible]

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The screenshot shows the AWS Management Console 'Launch Instance' wizard. The instance type is 't3.micro'. The key pair is 'MyKeyPair'. The network settings show the VPC as 'vpc-0a000000' and the subnet as 'subnet-0a000000'. The security group is 'sg-0a000000'. The storage settings show the root volume as 'gp3' and '16 GiB'. The 'Launch Instance' button is visible at the bottom right.

6. Launch. Confirm **no public IP**.

## Step 7: Create Application Load Balancer (ALB)

### 1) Open Load Balancers

1. AWS Console → **EC2** service → left nav → **Load Balancers** (under “Load Balancing”).
2. Click **Create Load Balancer** → choose **Application Load Balancer** → **Create**.

### 2) Configure basic settings (ALB)

1. **Name:** Web-ALB
2. **Scheme:** Internet-facing
3. **IP address type:** IPv4
4. **Listeners:** Keep default HTTP → **Port 80**
5. **Availability Zones / Subnets** (Network mapping):
  - VPC: ThreeTierVPC
  - Subnets: select one or preferably *two* or more public subnets (choose Public-Subnet and a second public subnet if you created one in another AZ). If you only created one public subnet, select it.
6. Click **Next: Configure Security Settings**. (For HTTP no TLS settings required — skip to next.)

### 3) Configure security groups

1. Select **ALB-SG** as the security group (it should allow inbound HTTP 80 from 0.0.0.0/0). If ALB-SG not listed, choose **Create new security group** and configure as in previous steps.
2. Click **Next: Configure Routing**.

### 4) Create target group

1. **Target group name:** Web-TG
2. **Target type:** Instance
3. **Protocol:** HTTP
4. **Port:** 80
5. **VPC:** ThreeTierVPC
6. Health checks:
  - Protocol: HTTP
  - Path: / (or /index.html) — default is fine
  - Advanced: leave defaults, or set Healthy threshold 2 / Unhealthy 2 / Interval 30s

The screenshot shows the AWS Management Console interface for creating a target group. The breadcrumb navigation at the top indicates the path: **EC2** > **Target groups** > **Create target group**. On the left, a progress bar shows three steps: **Step 1: Create target group** (active), **Step 2: Register targets**, and **Step 3: Register targets**. The main content area is titled **Create target group** and includes a sub-header: **Your load balancer routes requests to the targets in a target group and performs health checks on the targets.**

The **Basic configuration** section contains the following fields:

- Choose a target type:** Four radio buttons are present: **Instances** (selected), **IP addresses**, **Lambda function**, and **Application Load Balancer**. Each option has a list of bullet points describing its capabilities.
- Target group name:** A text input field containing **Web-TG**. A note below states: "A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen."
- Protocol:** A dropdown menu set to **HTTP**. A note states: "Protocol for load balancer-to-target communication. Can't be modified after creation."
- Port:** A text input field containing **80**. A note states: "Port number where targets receive traffic. Can be overridden for individual targets during registration."
- IP address type:** Two radio buttons: **IPv4** (selected) and **IPv6**. A note for IPv4 states: "Only targets with the indicated IP address type can be registered to this target group."
- VPC:** A dropdown menu set to **ThreeTierVPC**. A note states: "Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list."

The footer of the console shows the **CloudShell** icon, **Feedback** link, and copyright information: "© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Console preferences".

# Shaikh Ateeb Ahmed

## 25-08-2025

EC2 > Target groups > Create target group

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eni) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eni). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0ac41ed0700b54e18 (ThreeTierVPC)

10.0.0.0/16

Create VPC

Protocol version

☒ HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path

/

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

Up to 1024 characters allowed.

Advanced health check settings

Attributes

Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

Tags - optional

Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel

Next

7. Click **Next: Register Targets**.

### 5) Register targets (EC2 instances)

- In the **Register targets** pane, select both Web-1 and Web-2 (instances) and click **Add to registered**.
  - If your web servers are in different AZs, select both instances across AZs.
- After adding, click **Next: Review** → **Create**.

EC2 > Target groups > Web-TG > Register targets

Register targets

Select instances, specify ports, and add the instances to the list of pending targets. Repeat to add additional combinations of instances and ports to the list of pending targets. Once you are satisfied with your selections, click Register pending targets.

Available instances (4)

Filter instances

| Instance ID          | Name       | State   | Security groups | Zone        | Private IPv4 address | Subnet ID                | Launch time                         |
|----------------------|------------|---------|-----------------|-------------|----------------------|--------------------------|-------------------------------------|
| i-0511efa1d61229db76 | DB-Server  | Running | DB-SG           | ap-south-1c | 10.0.3.43            | subnet-0c0ff8ac5026b7d01 | October 21, 2025, 16:25 (UTC+05:30) |
| i-0b79f4e97bd663258  | App-Server | Running | App-SG          | ap-south-1b | 10.0.2.206           | subnet-0f0e0ad706640639e | October 21, 2025, 16:20 (UTC+05:30) |
| i-0b030a9f7a138b259  | Web-2      | Running | Web-SG          | ap-south-1a | 10.0.1.230           | subnet-061f986f07d46fc1  | October 21, 2025, 15:56 (UTC+05:30) |
| i-0b0561d3303d49f0b  | Web-1      | Running | Web-SG          | ap-south-1a | 10.0.1.51            | subnet-061f986f07d46fc1  | October 21, 2025, 15:48 (UTC+05:30) |

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-40535 (separate multiple ports with comma)

Include as pending below

2 selected are now pending below. Include more or register targets when ready.

Review targets

Targets (2)

Filter targets

| Instance ID          | Name  | Port | State   | Security groups | Zone        | Private IPv4 address | Subnet ID               | Launch time                         |
|----------------------|-------|------|---------|-----------------|-------------|----------------------|-------------------------|-------------------------------------|
| i-0b0f50a9f7a138b259 | Web-2 | 80   | Running | Web-SG          | ap-south-1a | 10.0.1.230           | subnet-061f986f07d46fc1 | October 21, 2025, 15:56 (UTC+05:30) |
| i-0b0561d3303d49f0b  | Web-1 | 80   | Running | Web-SG          | ap-south-1a | 10.0.1.51            | subnet-061f986f07d46fc1 | October 21, 2025, 15:48 (UTC+05:30) |

2 pending

Remove all pending

Register pending targets

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Account ID: 676b-292b-6d6c-9 Account ID: 676b-292b-6d6c-9

Asia Pacific (Mumbai) Shaikh Ateeb Ahmed

EC2 > Load balancers > Create Application Load Balancer

Create Application Load Balancer

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

Web-ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Internet-facing

Internal

Load balancer IP address type

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

IPv4

Dualstack

Dualstack without public IPv4

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and route within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups.

vpc-dac41ed700b54c18 (ThreeTierVPC)

Create VPC

IP pools

You can optionally choose to configure an IPAM pool as the preferred source for your load balancer IP addresses. Create or view pools in the Amazon VPC IP Address Manager console.

Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses, if the pool is deployed IPv4 addresses will be assigned by AWS.

Availability Zones and subnets

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

ap-south-1a (ap1-a21)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

Security groups

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can create a new security group.

Security groups

Select up to 3 security groups

ALB-SG

ap-011238d4d4642107 VPC: vpc-dac41ed700b54c18

Listeners and routing

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener: HTTP-80

Protocol

Port

80

Default action

Routing action

Forward to target groups

Redirect to URL

Return fixed response

Forward to target group

Choose a target group and specify routing weight or create target group.

Target group

Web-TG

Target type: Instance, IPv4 Target stickiness: Off

HTTP

Weight

1

Percent

100%

0.999

Add target group

You can add up to 4 more target groups.

Target group stickiness

Enables the load balancer to bind a user's session to a specific target. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

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Account ID: 676b-292b-6d6c-9 Account ID: 676b-292b-6d6c-9

Asia Pacific (Mumbai) Shaikh Ateeb Ahmed

EC2 &gt; Load balancers &gt; Create Application Load Balancer

Availability Zones and subnets

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

ap-south-1a (ap1-a21)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0c1798d073b46d14

Public Subnet

ap-south-1b (ap1-a23)

ap-south-1c (ap1-a22)

Security groups

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can create a new security group.

Security groups

Select up to 3 security groups

ALB-SG

ap-011238d4d4642107 VPC: vpc-dac41ed700b54c18

Listeners and routing

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener: HTTP-80

Protocol

Port

80

Default action

Routing action

Forward to target groups

Redirect to URL

Return fixed response

Forward to target group

Choose a target group and specify routing weight or create target group.

Target group

Web-TG

Target type: Instance, IPv4 Target stickiness: Off

HTTP

Weight

1

Percent

100%

0.999

Add target group

You can add up to 4 more target groups.

Target group stickiness

Enables the load balancer to bind a user's session to a specific target. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

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EC2

Load balancers

Create Application Load Balancer

Listeners and routing

Listener: HTTP80

Protocol

HTTP

Port

80

1-65535

Remove

Default action

Routing action

Forward to target groups

Redirect to URL

Return fixed response

Forward to target group

Choose a target group and specify routing weight or create target group

Target group

Web-TG

HTTP

Weight

1

0-999

Percent

100%

Add target group

You can add up to 4 more target groups.

Target group stickiness

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute stickiness.

Turn on target group stickiness

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

Add listener

You can add up to 45 more listeners.

Load balancer tags - optional

Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them. The 'Key' is required, but 'Value' is optional. For example, you can have Key = production-webserver, or Key = webserver, and Value = production.

Optimize with service integrations - optional

Optimize your load balancing architecture by integrating AWS services with this load balancer at launch. You can also add these and other services after your load balancer is created by revisiting the load balancer's "Integrations" tab.

CloudShell

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EC2

Load balancers

Create Application Load Balancer

Apply application layer security protections - in front of targets

Your choice of either a pre-defined security configuration with basic recommended AWS WAF security protections, or associate any of your existing WAF configurations for custom protections. [Additional charges apply](#)

Benefits and considerations

AWS Global Accelerator

Optimizes Performance, Availability

Apply global load balancing across multiple regions

Create an accelerator in your account with two global static IPs that act as a fixed entry point to your load balancer. If you do not need global static IPs or traffic management across multiple regions, select Amazon CloudFront. [Additional charges apply](#)

Benefits and considerations

Review

Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose **Create load balancer**.

Summary

Basic configuration

Name: Web-ALB

Scheme: Internet-facing

IP address type: IPv4

Network mapping

VPC: vpc-0bc41ed700b64d18

Public IPv4 IPAM pool: -

Availability Zones and subnets:

- ap-south-1a
  - subnet-0a37f9a9071a6-46f-1

Public Subnet

Security groups

ALB-SG

sg-09129f06cd463257

Listeners and routing

HTTP80 | Forward to 1 target group

Service integrations

Amazon CloudFront + AWS WAF + AWS WAF (WAF) -

AWS WAF -

AWS Global Accelerator -

Tags

-

Attributes

Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.

Creation workflow and status

Server-side tasks and status

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

Cancel

Create load balancer

CloudShell

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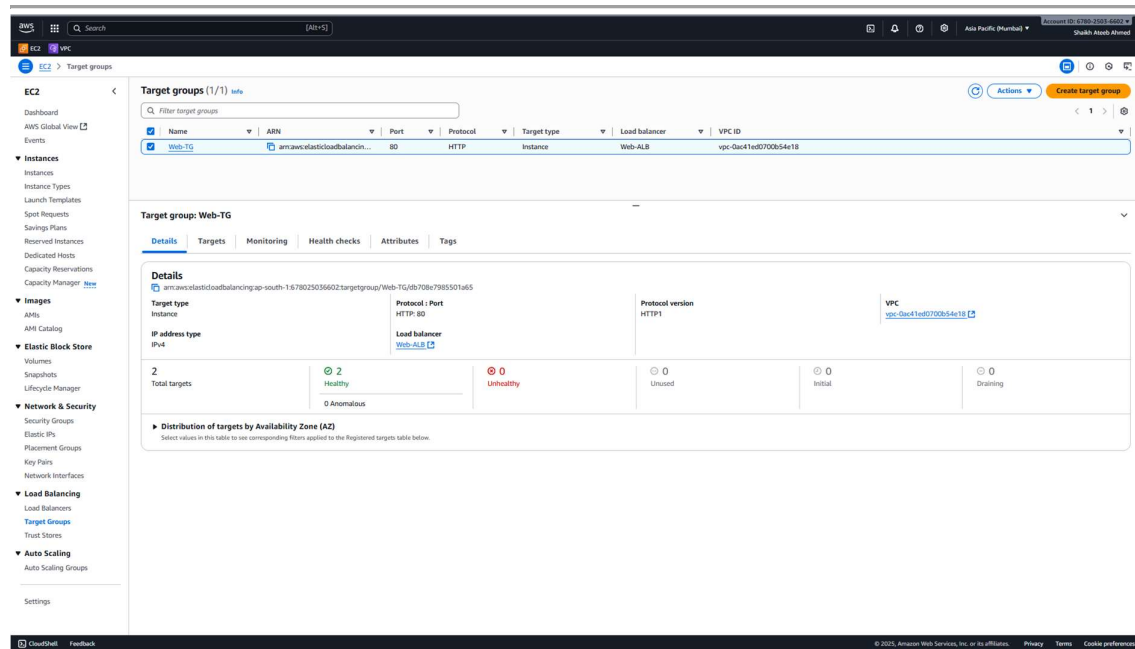
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# Shaikh Ateeb Ahmed

## 25-08-2025



## Step 8: Testing the full flow

### 1) Get ALB DNS name

- EC2 → Load Balancers → select Web-ALB → copy **DNS name** (Web-ALB-145371319.ap-south-1.elb.amazonaws.com).

### 2) Test ALB in a browser (basic check)

- Open a browser and paste the ALB DNS name. (after pasting click on the link and from https:// make it http:// Web-ALB-145371319.ap-south-1.elb.amazonaws.com)
- You should see the index page served by one of the web servers (the text you wrote with user-data).
- Refresh the page repeatedly you should see responses alternate between **Web Server 1** and **Web Server 2** (if the index messages differ). That demonstrates round-robin load balancing.



Shaikh Ateeb Ahmed  
25-08-2025

